

Assignment 1: The CNN Challenge

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Final result and validation protocol

The final model is an ImageNet-pretrained **ConvNeXt-Large** with all layers fine-tuned. It achieved **96.25% validation accuracy** and **96.28% validation macro F1**. After model selection was locked, a single labeled-test evaluation produced **98.50% accuracy** (394/400) and **98.48% macro F1**.

The 2,400 supplied training images were divided once using seed 0 into a fixed, class-stratified 1,920/480 train/validation split (120/30 images per class). SHA-256 grouping kept the six exact duplicate pairs, all in **LivingRoom**, on the same side of the split. An audit found no exact overlap with either test directory. The 400 labeled test images were excluded from every architecture, hyperparameter, and stopping decision; the 400 unlabeled **test2** images were unused. Mixed grayscale/RGB source files were all decoded as three-channel RGB for the final model.

The final recipe

Torchvision’s ConvNeXt-Large was initialized with **IMAGENET1K_V1** weights; the 1,000-way classifier was replaced by a 16-way head with dropout 0.3. Inputs were 224×224 RGB. Training augmentation used random resized crop (scale 0.75–1.0, ratio 0.8–1.25), horizontal flip ($p = 0.5$), brightness and contrast jitter (0.15), and random erasing ($p = 0.2$). Validation used resize then center crop. Both used ImageNet mean/std normalization.

All 196.25M parameters were optimized with AdamW (learning rate 10^{-4} , weight decay 10^{-4}), cosine decay, 0.1 label smoothing, batch size 32, AMP, and a 50-epoch budget with patience-10 validation early stopping. The best checkpoint occurred at epoch 2; training stopped after epoch 12. On an A100 40GB, the run took 202.3s and peaked at 9,364.6 MiB allocated GPU memory. The model has an estimated 34.36G MACs (68.72G FLOPs) per image; pooling and activation operations are excluded from that estimate.

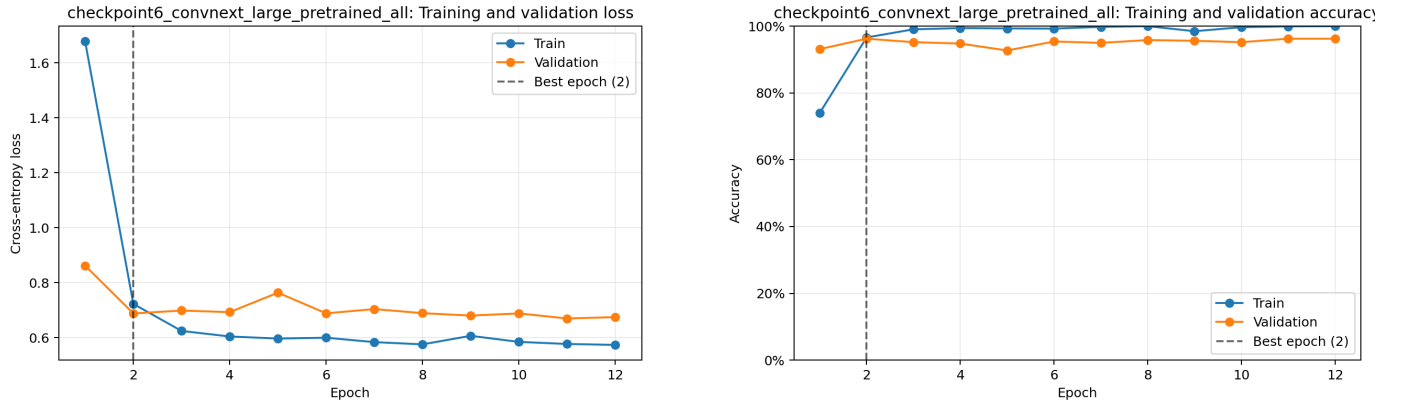


Figure 1: Checkpoint 6 learning curves. The saved checkpoint is chosen by validation accuracy, not the final epoch.

Experimental journey

Checkpoint	Controlled question / model	Val. acc.	Params	MACs/img
1	Supplied grayscale CNN	46.04%	0.058M	0.0006G
2	RGB augmentation + regularized scratch ResNet-18	76.46%	11.18M	0.59G
3	Scratch family winner: ResNeXt-50	78.33%	23.01M	1.38G*
4	ResNeXt-50, pretrained, last stage + head	94.79%	23.01M	4.23G
5	ConvNeXt-Tiny, pretrained, all layers	96.04%	27.83M	4.45G
5	EfficientNet-B0, pretrained, head only	93.33%	4.03M	0.38G
6	ConvNeXt-Large, pretrained, all layers	96.25%	196.25M	34.36G

*Checkpoint 3 used 128×128 inputs; checkpoints 4–6 used 224×224 .

The starter CNN’s widening train/validation gap motivated augmentation and regularization. Across scratch models, ResNeXt-50 led, but ImageNet pretraining was the dominant improvement: its four freezing strategies reached 94.17–94.79%, versus 82.50% from scratch at 224 pixels. The broader transfer matrix then selected fully fine-tuned ConvNeXt-Tiny. Scaling to Large gained only one validation image while using $7.05\times$ the parameters and $7.71\times$ the MACs. Large is therefore the final accuracy model, while Tiny is the stronger accuracy–efficiency choice. EfficientNet-B0 head-only is a notable low-training-cost alternative: 93.33% with only 20,496 trainable parameters.

Failure analysis and final test behavior

The clearest failure was scratch MobileNetV3-Large: validation remained at chance (6.25%) and every validation image was predicted as **Suburb**, although training accuracy reached 66.8%. This exposes a limitation of the “same recipe for every architecture” comparison: equal settings are controlled but not necessarily suitable. The failed run was retained rather than silently discarded; later comparisons used pretrained representations and scope-specific learning rates.

On the final test, 12/16 classes had 25/25 correct. **Kitchen**, **LivingRoom**, and **Mountain** each had 24/25; **TallBuilding** had 22/25, confused once each with **Forest**, **Industrial**, and **InsideCity**. The higher test than validation accuracy is plausible for these small samples and was not used to revise the model.

Reproducibility, checkpoint, and AI collaboration

Code and evidence are in the GitHub repository, tag `checkpoint6`. The exact configuration is `configs/checkpoint6/convnext_large_pretrained_all.yaml`; training is launched by `scripts/run_checkpoint6_convnext_large.sh`. The 749 MiB checkpoint is the release asset below, followed by its SHA-256:

```
checkpoint6_convnext_large_pretrained_all.pt
d011ab54be56d0fbb0c643c1f624d935b1521b1c21f2f73f3df5b2f4649f8919
```

Saved histories, validation matrices, complexity measures, and the immutable test JSON are tracked in `results/`.

OpenAI Codex helped inspect the handout/notebook, audit duplicates and color modes, implement reusable training/evaluation/plotting infrastructure, and organize evidence. The student ran every experiment, reviewed outputs, chose the hypothesis sequence and final model, and retained responsibility for the interpretation. One ineffective AI-assisted choice—using a shared scratch recipe for MobileNet—is reported above. Full disclosure is in `AI_USAGE.md`.